

Integrated Results Management System (IRMS)

Zonal Meeting Presentation – ACSEE Operations, Governance, and Roadmap

System Presentation Team

PMO-RALG | Examination Operations

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Presentation Objectives

- Give a complete, shared understanding of what IRMS delivers end-to-end.
- Explain how ACSEE registration, mark entry, moderation, locking, compute, publication, and evaluation work together.
- Show system strengths for operational control, accountability, and speed.
- Highlight current live status (data and processing snapshot).
- Align zonal decisions on scale-up priorities and risk mitigation.

Why IRMS Matters at Zonal Level

Operational Challenge

Different actors (region, district, school) need one controlled workflow for accurate, timely, and auditable results.

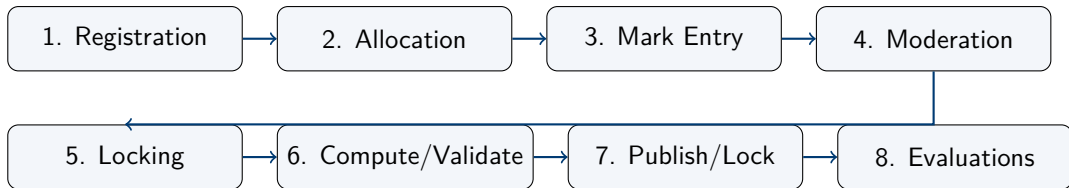
Without IRMS

- Fragmented files and manual reconciliation
- Slow incident tracing
- Weak audit defensibility
- Delayed publication decisions

With IRMS

- Single lifecycle from registration to publication
- Validation and moderation gates
- Immutable operational audit trails
- Faster readiness and governance decisions

IRMS End-to-End Service Scope



Principle: Every stage has explicit controls before moving to the next stage.

Technology and Platform Architecture

Technology Baseline

- Laravel 12.48.1, PHP 8.3
- Blade + Alpine.js front-end
- Tailwind CSS 4, Vite build pipeline
- SQLite operational database
- Filament v3 for admin resources

Architecture Style

- CRUD modules for master/reference data
- Workflow modules for lifecycle-heavy operations
- Policy + gate enforcement for sensitive actions
- Audit-log oriented governance model

Design Outcome

Balance between operational speed (workflow pages) and maintainability (admin resources and policy-driven control).

Core Functional Modules (What Zones Use)

Module	Scope and Value
Registration	Candidate, school, district, and exam-year enrollment with scope control.
ACSEE Allocation	Combination/subject allocation and private-candidate subject assignment workflows.
Mark Entry	CSV/ZIP ingestion, validation, batch lifecycle, moderation, approval, lock/unlock.
Results Processing	Compute/validate, grading profile control, publish/lock snapshots, admin unlock.
Evaluations	School/council/district/region views, rankings, outlier/extremity analytics.
Governance & Backup	Authentication auditing, action trails, backup/restore with restore audit logs.

ACSEE Mark Entry Workflow (Operational View)

- 1 Select context (exam year, region, district, school, subject).
- 2 Download template (paper structure and format constraints).
- 3 Upload marks (CSV/ZIP) into draft batch.
- 4 Run validation (candidate link, subject registration, paper completeness, ranges).
- 5 Send for moderation; resolve flagged errors/incidents.
- 6 Approve and lock batch (immutable stage for compute readiness).

Control Principle

No progression to compute stage without passing lifecycle gates and data-quality checks.

Data Quality and Compute Rules

- Subject marks normalized to 100 using configured paper weights.
- Compute policy rounds normalized subject marks to whole numbers.
- Canonical paper maxima: Paper 1/2 = 100, Paper 3 = 50 (unless configured).
- Irregular statuses handled explicitly: INC, ABS, X with priority logic.
- Readiness indicators depend on promoted/valid marks, not only uploaded rows.

Operational Benefit

Stronger consistency across schools and councils; fewer disputes during publication and post-publication review.

Access Control

- Custom RBAC with role + scope binding
- Region/district/school isolation
- Policy and gate checks for critical actions
- Password-change enforcement and status control

Governance Controls

- Authentication and action audit logging
- Moderation and unlock actions are recorded
- Publish/lock and admin-unlock events traceable
- Backup and restore actions audited

Results Governance: Publish, Lock, Unlock

- Compute runs generate versioned results artifacts.
- Publication is separated from locking to support governance review.
- Locking protects final output from accidental mutation.
- Admin unlock exists for controlled exception handling (with reason trail).
- Snapshot/version workflow supports defensible result history.

Leadership Value

Decisions are reversible only through governed pathways, not ad-hoc edits.

Live Operational Snapshot (Database, Feb 27, 2026)

Indicator	Current Count
Regions in scope	8
Districts in scope	19
Registered schools	93
Candidate master records	11,168
ACSEE registrations (2026)	11,168
ACSEE subject selections (2026)	48,299
ACSEE mark-import batches (2026 to- tal)	581
Locked ACSEE batches (2026)	551
Active result runs (pending/in_progress)	0
Latest result run (ID 30)	completed, 8,973/8,973 processed

Source: local production-aligned database snapshot generated at 2026-02-27 03:26.

System Strengths for Zonal Operations

- ① **Traceability:** full lifecycle visibility from upload to publication.
- ② **Control:** moderation, lock, unlock, and publication governance.
- ③ **Data quality:** strict validation and irregularity handling.
- ④ **Scalability of operations:** bulk import and scope-based filtering.
- ⑤ **Transparency:** public and internal result views with aligned logic.
- ⑥ **Audit readiness:** defensible logs for compliance and incident response.

Key Risks and Improvement Priorities

- **Role-system consistency:** continue hardening one authorization model across all controllers.
- **Operational resilience:** strengthen stale-run auto-recovery and queue/worker observability.
- **Database growth path:** plan migration path for higher write concurrency when scale rises.
- **UI clarity:** continue reducing ambiguity in lifecycle labels and readiness counters.
- **Institutional readiness:** keep zonal training and SOP updates synchronized with releases.

Evidence source includes architecture and lifecycle audit documentation in repository.

90-Day Action Plan (Zonal Focus)

Window	Priority	Expected Outcome
0–30 days	Lifecycle reliability hardening	Fewer stuck runs, faster operator recovery, clearer readiness status.
31–60 days	Authorization consistency pass	Reduced permission edge cases and stronger governance confidence.
61–90 days	Scale and reporting optimization	Better performance under peak loads and improved zonal monitoring dashboards.

Execution Model

Joint workstream: Zonal operations + system team + governance focal persons.

What We Need From Zonal Leadership

- Confirm standard operating procedures for exception handling (unlock/recompute/escalation).
- Nominate zonal super-users for rapid support and training-of-trainers.
- Approve periodic operational review cadence (weekly in exam season).
- Endorse phased hardening priorities and milestone tracking.

IRMS is ready for controlled scale-up with governance-first execution.

Appendix: Source Artifacts Used

- `docs/IRMS_ADMIN_PANEL_ARCHITECTURE_REPORT.md`
- `MARK_ENTRY_WORKFLOWS.md`
- `COMPLETE_SYSTEM_SUMMARY.md`
- `app/Http/Controllers/Api/Results/AcseeLifecycleApiController.php`
- Live DB snapshot queries executed via `php artisan tinker` on 2026-02-27.

Note: Update the snapshot slide before final delivery if new data is loaded.

Thank You

Questions & Discussion